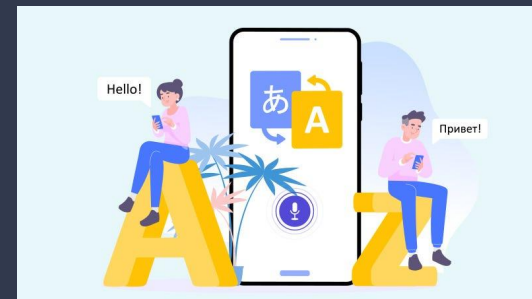


Phrase-Based & Neural Unsupervised Machine Translation



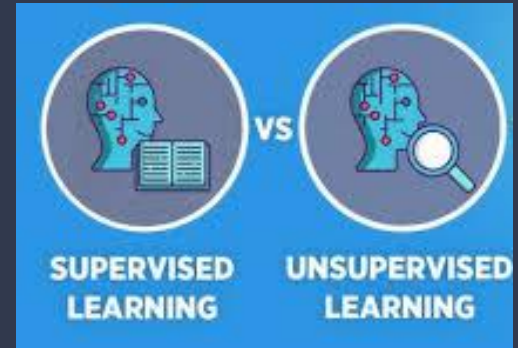
Prof. Laura Nenzi
Nooshin Behrooz
July 2026



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Table of Contents

1. *Introduction: The Problem*



1. Introduction: The Problem

Before 2017/2018 (around publishing this paper)

People believed: No parallel data → No translation system

Translation was considered almost impossible without bilingual sentence pairs.

Why? Because MT Systems, which are very effective,
but their success strongly **relies on large amounts of parallel sentences.**

1. Introduction: The Problem

1) Why does MT need parallel corpora?

MT learns by matching sentence pairs across languages to learn word, phrase, and sentence alignments.

Examples of Supervised MT systems:

Neural Machine Translation (NMT) (Encoder–Decoder models)

- Seq2Seq LSTM (2014)
- Transformer (2017)
- T5 (2019/2020)
- mBART (2020)
- NLLB (2022)

Statistical Machine Translation (SMT)

- Phrase-Based Statistical Machine Translation (PBSMT)
- Moses (popular PBSMT toolkit)

2) Why is this a problem?

Many language pairs **lack sufficient parallel corpora**, especially **low-resource languages**.

- French–Urdu
- Minority languages

Situation	Expected Performance
Lots of direct LA ↔ LB	Excellent
No Di LA ↔ LB Data, but lots of En links	Good
Very little data for both languages	Weak

1. *Introduction: The Solution*

The paper introduces:

Phrase-Based & Neural Unsupervised Machine Translation

Goal:

translating between two languages **without parallel sentence pairs**
using only **monolingual corpora** in each language

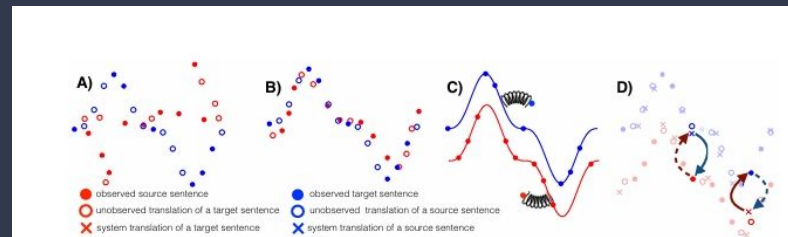
How?

2. The Three key Principles

2.1 Initialization

2.2 Language Modeling

2.3 Iterative Back-Translation



2. The Three key Principles

2.1 Initialization

Why is it needed?

There is **no parallel data**, so the model **needs a weak starting bridge** between the two languages

Main idea:

The system **starts** with an approximate **cross-lingual alignment**, so training does **not begin from pure randomness**.

In NMT:

[One Shared Embeddings] from both monolingual corpora

In PBSMT:

An **Initial Phrase Table** is built from a bilingual dictionary inferred from cross-lingual embeddings

Why is it important?

Bad Initialization can send the system in the **Wrong Direction**, and **Back-Translation** may not fully fix it

2. The Three key Principles

2.2 Language Modeling

Why is it needed?

After initialization, **translations are still noisy and unnatural**

Main idea:

Train a language model on each language separately, so the system **learns what a correct sentence looks like**, the **Structure** and **Fluency of each language**, and acting as a correction mechanism for noisy outputs.

In NMT:

Use a **Denoising Autoencoder** (real sentence → add noise → reconstruct clean sentence)

In PBSMT:

Use an **n-gram language model** (such as KenLM, to score and prefer fluent translations)

2. The Three key Principles

2.3 Iterative Back-Translation

Why is it needed?

Even with initialization and language modeling, there are **still no parallel sentence pairs**

Main idea:

Automatically generate pseudo-parallel data by translating sentences between the two languages

How does it work?

1. Translate a sentence to the other language.
2. Translate it back.
3. Compare with the original sentence.
4. Update the model.

Why iterative?

- Early iterations → poor translations
- Later iterations → better synthetic data
- Final iterations → stronger translation models

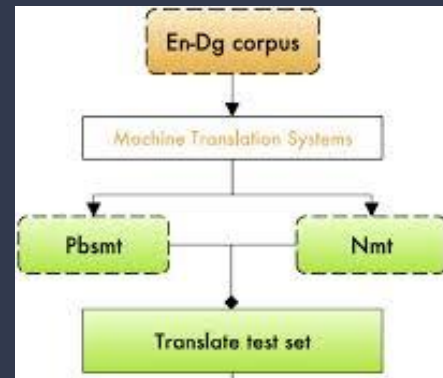
Key idea:

Back-translation turns an **unsupervised problem** into a **supervised learning problem** using automatically generated sentence pairs

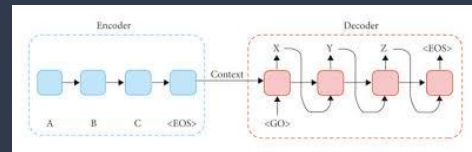
3. The Two Models

3.1 Neural Machine Translation-NMT

3.2 Phrase-Based Statistical Machine Translation-PBSMT



3. The Two Models: 3.1 Neural Machine Translation (NMT)



1. Initialization

Goal: Create a **Shared Embedding Space** for English and Italian **without parallel sentences**

Step 1: Combine the two monolingual corpora

English corpus + Italian corpus

Step 2: Apply joint *BPE tokenization

BPE splits words into smaller subword units, **Ex:** information / informazione → inform + ation / inform + azione

Step 3: Build one shared BPE vocabulary

Both languages use the same subword_vocabulary

inform, ation, azione,

Step 4: Use **One Shared Embedding Table**

The same embedding layer is used for English and Italian subwords

inform → [0.12, -0.30, 0.55, ...]

ation → [0.44, 0.18, -0.09, ...]

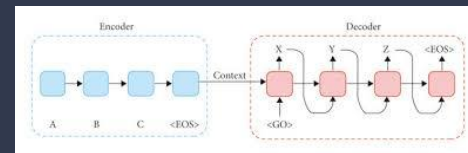
azione → [0.40, 0.22, -0.11, ...]

Step 5: Share encoder and decoder parameters

The model learns both languages in one common NMT architecture

***Byte Pair Encoding**

3. The Two Models: 3.1 NMT



2. Language Modeling (Denoising Autoencoder)

Goal: Learn grammar, word order and fluency of each language separately using monolingual corpora.

ENCODER

INPUT: Corrupted Sentence_En: I l ve pizz

↓
Embedding → Self-Attention → Feed Forward
(initialized by **Shared BPE Embeddings**)

↓
OUTPUT: Hidden States ($B \times N \times H$)

DECODER

INPUT: Clean Sentence_En (Teacher Forcing): <START> I love pizza

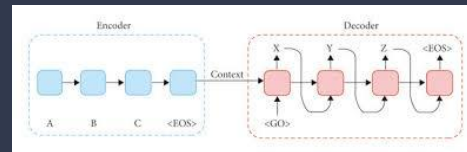
↓
Decoder Hidden States → **Cross Attention** with **Encoder Hidden States**

↓
OUTPUT: Prediction Matrix ($B \times N \times V$), Ex: I like pizza

LOSS_LM_En: Prediction vs Original Clean Sentence
Repeat for Italian corpus

IR relation: **Cross-attention** performs **soft neural retrieval**, where each decoder query scores and retrieves relevant encoder hidden states to generate the next translation token. ($Q \times K \rightarrow \text{scores} \rightarrow \text{softmax} \rightarrow \text{weights} \rightarrow \text{weighted sum of } V$)

3. The Two Models: 3.1 NMT



3. Iterative Back-Translation

Goal: Create pseudo-parallel sentence pairs from monolingual corpora and learn translation without parallel data.

Step 1: Generate a Translation ENCODER

INPUT: **Clean Sentence_En**, Ex: I love pizza
OUTPUT: **Hidden States_cl-sent**



DECODER

INPUT: <IT> <START>
OUTPUT: Predict **Synthetic Translation_It**, Ex: Io Preferisco pizza

Step 2: Train Reverse Direction ENCODER

INPUT: **Synthetic Translation_It**, Ex: Io Preferisco pizza
OUTPUT: **Hidden States_syn-tr**



DECODER

INPUT: **Clean Sentence_En**: <START> I love pizza
(Teacher Forcing)
OUTPUT: **Predict Sentence_En**, Ex: I like pizza

Step 3: Back-Translation Loss

LOSS_BT-En: Prediction vs Original Clean Sentence : I like pizza vs I love pizza

Step 4: Repeat in Opposite Direction: Italian → English

LOSS_total = $L_{LM_En} + L_{LM_It} + L_{BT_En} + L_{BT_It}$ → **Update:** Encoder / Decoder / Embedding Parameters

3. The Two Models: 3.2 Phrase-Based Statistical Machine Translation

1. Initialization



Goal: Build an **Initial Phrase Table** for source (En) $\leftrightarrow$ target (It) **without parallel sentences**

Step 1: Tokenization English/Italian

Step 2: Train monolingual embeddings separately for both languages : En/It corpus $\rightarrow$ [En/It Embeddings Vector]

Step 3: Align the **two embedding spaces** using a **rotation matrix W^***

$W(\text{Emb_En}) \approx \text{Embedding_It}$: $W(\text{love}) \approx \text{amare}$

Step 4: Use Cosine similarity + Nearest-neighbor retrieval to find candidate translations

Cos ($W(\text{love})$, amare)

love $\rightarrow$ amare 0.92

love $\rightarrow$ piacere 0.71

love $\rightarrow$ adorare 0.15

....

love $\rightarrow$ casa 0.01

.....

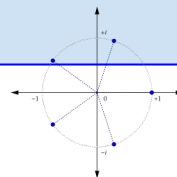
Build a temporary bilingual lexicon by Choosing the **Nearest-Neighbor Retrieval (top_k selection)**

Step 5: Refine W and Repeat to reach **final W**

(Use the temporary lexicon to improve the alignment)

***Isomorphism Hypothesis:** The **geometric structures** of **language spaces** share **similar topologies**.

Calculation W , Gradient-based methods or SVD (Singular Value Decomposition)



En: man + crown = king
It: uomo + corona = re

3. The Two Models: 3.2 Phrase-Based Statistical Machine Translation

1. Initialization



Step 6: Build an Initial Bilingual Dictionary

Step 7: Convert Similarities into Probabilities: $\sum P(It_w | love) = 1$



Initial Phrase Table

Initial Bilingual Dictionary

English	Italian	Cos_sim
love	amare	0.92
love	piacere	0.71
love	adorare	0.15
good morning	buongiorno	0.95
good morning	buona mattina	0.45



Initial Phrase Table

English	Italian	Prob
love	amare	0.55
love	piacere	0.42
love	adorare	0.03
good morning	buongiorno	0.75
good morning	buona mattina	0.25

3. The Two Models: 3.2 PBSMT

2. Language Modeling



Goal: Learn **fluent** Language (It/En) sentences from **monolingual corpora** and **Score** translation candidates

Step 1: Train a Statistical n-gram Language Model for each language

Stores n-gram statistics: how often words follow each other

$P(\text{buongiorno}) \times P(\text{ragazzi} | \text{buongiorno}) > P(\text{mattina}) \times P(\text{ragazzi} | \text{mattina})$

Step 2: Decoder Generates Translation Candidates from Phrase Table

good morning guys → buona mattina ragazzi, buongiorno ragazzi, ragazzi buongiorno

Step 3: Score Candidate Translations

Language Model → Candidate Fluency_Scores

(as it has learned **common word sequences** from corpora)

Reordering Model → Word Order_Scores

Step 4: Decoder (search Algorithm) Selects The Best Translation

Decoder Score = Log(Phrase Translation Score) + Log(LM_Fluency Score) + Log(W_Order_Score)

Decoder Winner: good morning guys → **buongiorno ragazzi**

Candidate_Trans	LM_Fluency_Score
buona mattina ragazzi	Low
buongiorno ragazzi	Higher

Candidate_Trans	W_Order_Score
ragazzi buongiorno	Low
buongiorno ragazzi	Higher

3. The Two Models: 3.2 PBSMT

3. Back-Translation



Goal: Improve translation quality by creating synthetic parallel corpora and updating the Phrase Table

Step 1: Build Synthetic Parallel Corpora

(Using the Level 2 PBSMT Translator)

1.1 Translate EN corpus → IT and create Synthetic Corpus A

1.2 Translate IT corpus → EN and create Synthetic Corpus B

(Called **synthetic parallel corpora** because one side is original and the other side is machine-generated)

Step 2: Re-Estimate the Phrase Table → Update phrase table

(So the phrase table becomes better)

Step 3: Build Better Translators

(Decoder translates using the updated Phrase Table)

Step 4: Repeat Iteratively step 1 → 3

Synthetic Corpus A

English	Synthetic Italian
good morning guys	buongiorno ragazzi

Synthetic Corpus B

Italian	Synthetic English
buongiorno tutti	good morning all

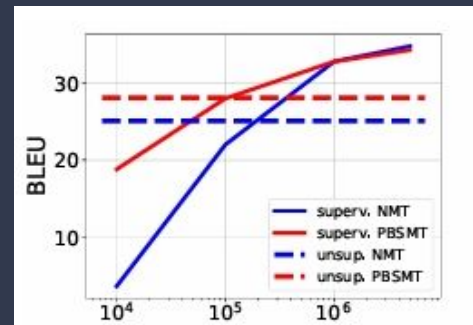
Updated Phrase Table

English	Italian	Prob
good morning	buongiorno	0.75 → 0.95↑
good morning	buona mattina	0.25 → 0.05↓

4. The Results and Comparison

4.1 Paper Results

4.2 Our Implementation Results



4.1 Paper Results: Table 2

1st Result: Proposed Unsupervised MT **Vs Previous Unsupervised MT**

Paper model	*BLEU for en-fr
Paper_NMT	25.1 vs 17.0
Paper_PBSMT	28.1 vs 17.0
Paper_Hybrid	27.6 vs 17.0

NMT

All paper models **outperform** previous unsupervised NMT.

Model	en-fr	fr-en	en-de	de-en
(Artetxe et al., 2018)	15.1	15.6	-	-
(Lample et al., 2018)	15.0	14.3	9.6	13.3
(Yang et al., 2018)	17.0	15.6	10.9	14.6
NMT (LSTM)	24.5	23.7	14.7	19.6
NMT (Transformer)	25.1	24.2	17.2	21.0
PBSMT (Iter. 0)	16.2	17.5	11.0	15.6
PBSMT (Iter. n)	28.1	27.2	17.9	22.9
NMT + PBSMT	27.1	26.3	17.5	22.1
PBSMT + NMT	27.6	27.7	20.2	25.2

2nd Result: Proposed NMT **Vs PBSMT** **Vs HYBRID Model**

Table 2: Comparison with previous approaches

1. **PBSMT**: Surprisingly **stronger than NMT**!
2. ***PBSMT + NMT**: The **Best Overall** big-picture model, **Especially for harder pairs**.

***BLEU**: is an **automatic MT evaluation score** that measures how similar a machine translation is to a human translation, by comparing n-gram overlap between machine and human translation.

***PBSMT + NMT**: Synthetic parallel sentences → NMT, back-translation step

Language pair	Dataset
English ↔ French	WMT'14
English ↔ German	WMT'16

4.1 Paper Results: Table 3

1st Result: The effect of the number of Back-Translation Iterations in PBSMT

Table 3 shows that **PBSMT improves** as the **number of iterations increases**

2nd Result: Final PBSMT Vs NMT Vs HYBRID Model for more datasets

NMT: strong, but weaker for harder pairs

PBSMT: stronger and more stable than NMT

***PBSMT + NMT: Best overall** big-picture mode (Especially for harder pairs.)

	en → fr	fr → en	en → de	de → en	en → ro	ro → en	en → ru	ru → en
<i>Unsupervised PBSMT</i>								
Unsupervised phrase table	-	17.50	-	15.63	-	14.10	-	8.08
Back-translation - Iter. 1	24.79	26.16	15.92	22.43	18.21	21.49	11.04	15.16
Back-translation - Iter. 2	27.32	26.80	17.65	22.85	20.61	22.52	12.87	16.42
Back-translation - Iter. 3	27.77	26.93	17.94	22.87	21.18	22.99	13.13	16.52
Back-translation - Iter. 4	27.84	27.20	17.77	22.68	21.33	23.01	13.37	16.62
Back-translation - Iter. 5	28.11	27.16	-	-	-	-	-	-
<i>Unsupervised NMT</i>								
LSTM	24.48	23.74	14.71	19.60	-	-	-	-
Transformer	25.14	24.18	17.16	21.00	21.18	19.44	7.98	9.09
<i>Phrase-based + Neural network</i>								
NMT + PBSMT	27.12	26.29	17.52	22.06	21.95	23.73	10.14	12.62
PBSMT + NMT	27.60	27.68	20.23	25.19	25.13	23.90	13.76	16.62

Table 3: Fully unsupervised results

Table 3 confirms the results of Table 2 for more dataset, that **combining PBSMT with NMT** (Not inverse) **often gives the strongest overall unsupervised MT system**

Language pair	Dataset
English↔Romanian	News Crawl + WMT'16 Romanian monolingual data
English↔Russian	WMT monolingual News Crawl 2007–2017

4.1 Paper Results: figure 2

Unsupervised Models Competes with Supervised Models

- When **parallel data is small (low-resource languages)**, the **unsupervised models** can **outperform** supervised models.
- **Unsupervised PBSMT** achieves **similar performance** to **supervised PBSMT** trained with around **100,000 parallel sentences**.
- But when the **supervised model has a large parallel corpus**, especially around **1000,000** sentence pairs, supervised MT **becomes stronger**.

Examples:

Case	Parallel Data Used By Supervised	Supervised BLEU	UnSupervised BLEU	
Romanian → English	6,000	22.9	23.9	Outperforming UnSupervised method by 1 point!
Urdu → English	800,000 noisy data	9.8	12.3	both low resource and distant

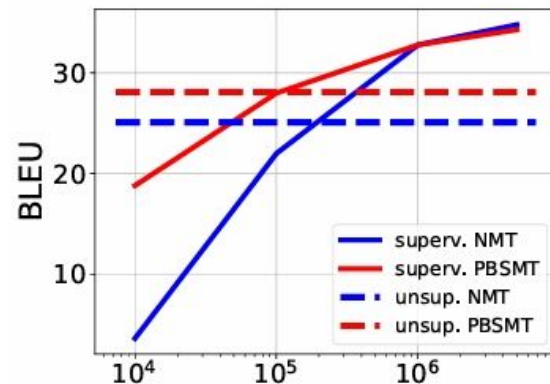


Figure 2: Comparison between **supervised** and **unsupervised** approaches on WMT'14 En-Fr, as we vary the **number of parallel sentences** for the **supervised methods**.

4.2 Our Implementation Results: Toy NMT

Goal: Implement a **light version** of the paper's NMT idea on a small **English–Italian** dataset.

Part	Original Paper NMT	Our Toy Non-Parallel NMT
Data	Large monolingual corpora; no parallel sentence pairs for training.	10,000 English sentences and 10,000 Italian sentences from different OPUS Books ranges; parallel training pairs used: 0.
Initialization	Joint BPE tokenization, shared BPE vocabulary, shared embeddings, and shared encoder/decoder parameters.	Simple tokenization, one shared word-level vocabulary from both monolingual corpora, one shared embedding layer, and one Transformer model.
Architecture	Full NMT architecture in the paper, including LSTM/Transformer variants.	Small Transformer encoder-decoder: d_model=192, 4 heads, 2 encoder layers, 2 decoder layers.
Level 1	Initialization gives the model a shared cross-lingual starting space.	Monolingual autoencoder warm-up: English -> English and Italian -> Italian. No EN-IT supervised warm-up.
Level 2	Denoising autoencoder learns fluency and structure from monolingual sentences.	DAE loss: corrupted English -> clean English and corrupted Italian -> clean Italian.
Level 3	Iterative back-translation creates synthetic parallel data in both directions.	Light bidirectional BT: EN -> synthetic IT -> reconstruct EN, and IT -> synthetic EN -> reconstruct IT.
Training schedule	Language modeling and back-translation are trained iteratively in the full system.	Level 2 and Level 3 are trained jointly in one loop using one combined loss: LM_EN + LM_IT + BT_EN + BT_IT.
Back-translation size	Uses large-scale monolingual data and several iterations.	Uses 1,500 sentences for light back-translation to keep the experiment fast.
Evaluation	BLEU on standard MT benchmarks.	BLEU on 50, 100, 250, and 500 held-out OPUS Books sentence pairs. Parallel data is used only for evaluation.

4.2 Our Implementation Results: Toy NMT

Model Setup:

1. **Small supervised warm-up:** learns a basic EN $\leftrightarrow$ IT translation mapping with one shared embedding layer for both languages.
2. **Denoising autoencoder:** learns fluency from noisy monolingual sentences
3. **Light back-translation:** creates synthetic sentence pairs and retrain the model

Evaluation:

Number of Sentences	BLEU
50	0.07
100	0.12
250	0.08
500	0.06

Training data	10,000 En sentences 10,000 IT sentences from OPUS Book
back-translation-Synthetic Data	1,500 sentences

4.2 Our Implementation Results: Toy PBSMT

Goal: Implement a **light version** of the paper's PBSMT idea on a small **English–Italian** dataset.

Part	Original Paper PBSMT	Our Toy PBSMT
Data	Large monolingual corpora; no parallel sentence pairs.	Small English-Italian dataset from OPUS Books.
Initialization	Cross-lingual embeddings, inferred bilingual dictionary, and initial phrase table.	Word2Vec embeddings, rotation matrix W , nearest-neighbor retrieval, and initial phrase table.
Embedding alignment	Robust cross-lingual embedding alignment.	Simplified alignment using <code>orthogonal_procrustes()</code> .
Phrase table	Full PBSMT phrase table with translation probabilities.	Simplified phrase table from Word2Vec candidates plus IBM1 toy phrase extraction.
Language model	KenLM n-gram language model.	Simple bigram language model implemented with <code>Counter()</code> .
Decoder	Full PBSMT decoder.	Toy decoder with beam search, phrase score, LM score, and simple reordering.
Back-translation	Multiple iterative back-translation rounds.	One bidirectional back-translation iteration for speed.
Evaluation	BLEU on standard MT benchmarks.	BLEU on 50, 100, 250, and 500 test sentences.

4.2 Our Implementation Results: Toy PBSMT

Model Setup:

1. **Initialization:** builds an initial EN $\leftrightarrow$ IT phrase table using aligned **Word2Vec** embeddings and **nearest-neighbor retrieval**.
2. **Language modeling + decoding:** uses a **Bigram language** model to score fluent Italian translation candidates.
3. **Bidirectional back-translation:** creates **synthetic EN $\rightarrow$ IT and IT $\rightarrow$ EN pairs** and updates the phrase table.

Evaluation Results:

Number of Sentences	BLEU
50	0.19
100	0.17
250	0.19
500	0.17

Training data	10,000 En sentences 10,000 IT sentences from OPUS Book
back-translation-Synthetic Data	800 sentences

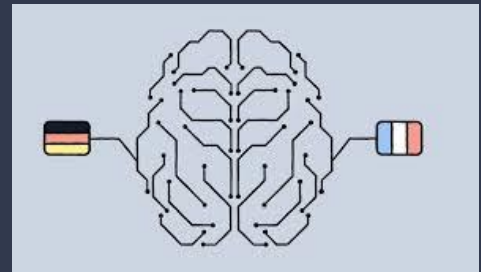
Result: Toy_PBSMT gives **higher BLEU** than the Toy_NMT.

4.2 Our Implementation Results: Toy PBSMT

Toy PBSMT Sample Translations

#	English Source	Reference Italian	Toy PBSMT Output
0	Source: Project Gutenberg	Source: www.liberliber.it/Audiobook available ...	polso: macchine gutenberg
1	Jane Eyre	Jane Eyre	iddio sta
2	Charlotte Brontë	Charlotte Brontë	charlotte bronte
3	CHAPTER I	PARTE PRIMA	circolare i
4	There was no possibility of taking a walk that...	I. In quel giorno era impossibile passeggiare.	dare andata no ostilità fu velo a futuro invec...
5	We had been wandering, indeed, in the leafless...	La mattina avevamo errato per un'ora nel bosch...	li tornato tuškevic posta, probabilmente, in e...
6	I was glad of it: I never liked long walks, es...	Ne ero contenta. Non mi sono mai piaciute le l...	i andata sapere fu neanche: i niente venire to...
7	The said Eliza, John, and Georgiana were now c...	Eliza, John e Georgiana erano aggruppati in sa...	espose pensò vedevano, sorprendente, cura qual...
8	Me, she had dispensed from joining the group; ...	Ella mi aveva proibito di unirmi al loro grupp...	me, cioè tornato dispensed in falci espose avv...
9	"What does Bessie say I have done?" I asked.	— Che cosa vi ha detto Bessie di nuovo sul con...	" utile vuoi vede ero i anch parla? " i ».

5. Conclusion



5. Conclusion

The contribution of this **paper** is that it **combines** and **improves 3 principles** in a unified framework for **Unsupervised NMT** and **PBSMT**.

- **Initialization** (shared embeddings/initial phrase table)
- **Language modeling**
- **Iterative bidirectional back-translation**

Result 1: The **proposed unsupervised MT** models strongly **outperform** **previous unsupervised MT** methods.

Result 2: from table 2 & 3 we saw **PBSMT** is surprisingly **Stronger** than **NMT** in this setting, while **PBSMT + NMT** gives the **Best overall unsupervised MT** system (Especially for harder pairs).

Result 3: When **parallel data is small** (**low-resource languages**), the **unsupervised models** can **outperform** supervised models

Result 5: Our **Toy PBSMT** achieves **higher BLEU** than our **Toy NMT**, which is **consistent** with the **paper's finding** that **PBSMT** can **outperform NMT** in low-resource unsupervised settings.

Number of Sentences	Non-Parallel Toy NMT BLEU	Toy PBSMT BLEU (After BT)
50	0.07	0.19
100	0.12	0.17
250	0.08	0.19
500	0.06	0.17

Reference

Phrase-Based & Neural Unsupervised Machine Translation

Guillaume Lample, Myle Ott, Alexis Conneau, Ludovic Denoyer, Marc'Aurelio Ranzato

arXiv:1804.07755v2

Thank you so much for your attention

